

# Nested Loops

# Nested loop

## Key points of the structure:

- We write a brand new loop inside the body of another loop.
- In the nested loop we need to make sure to use different variable than the one we used in the outer loop.
- For each iteration of the outer loop, the inner loop executes all its iteration.

## Structure of **nested for loop**:

```
1. for ( INITIALIZATION_1 ; CONDITION_1 ; STEPS_1 ) {  
2.     for ( INITIALIZATION_2 ; CONDITION_2 ; STEPS_2 ) {  
3.         BODY OF THE LOOP  
4.     }  
5. }
```

# Nested loop (contd.)

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

Here, i is being initialized with the value 0.

So, **i = 0**

# Nested loop (contd.)

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

Now, j is being initialized with the value 0 as well.

So, **i = 0 & j = 0**

# Nested loop (contd.)

## OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

Now, i and j are being printed

So, **i = 0 & j = 0**

# Nested loop (contd.)

OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.         System.out.println("nested ended");
8.     }
9. }
10. }
```

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> run NestedLoop1

0 0

We've reached the end of the *inner loop* we have to move up again and increment j then check condition.

So, **i = 0 & j = 0**

# Nested loop (contd.)

## OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

Incrementing the value of j by 1 we get, **j = 1** and **j < 3** is true.

So, **i = 0 & j = 1**

# Nested loop (contd.)

## OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

Now, i and j are being printed

So, **i = 0 & j = 1**

# Nested loop (contd.)

## OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

Incrementing the value of j by 1 we get, **j = 2** and **j < 3** is true.

So, **i = 0 & j = 2**

# Nested loop (contd.)

OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

Now, i and j are being printed

So, **i = 0 & j = 2**

# Nested loop (contd.)

OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

We've reached the end of the *inner loop* we have to move up again and increment j then check condition.

So, **i = 0 & j = 2**

# Nested loop (contd.)

OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

Incrementing the value of j by 1 we get, **j = 3** and **j < 3** is **false**.

Now, the nested loop will end and **j will be destroyed**.

So, **i = 0 & j = 3**

# Nested loop (contd.)

OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

We've reached the end of the *inner loop* we have to move up again and increment j then check condition.

So, **i = 0 & j = 1**

# Nested loop (contd.)

## OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

nested ended

Now, the print statement will be executed.

# Nested loop (contd.)

OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

nested ended

We've reached the end of the *outer loop* we have to move up again and increment *i* then check condition.  
So, **i = 0**

# Nested loop (contd.)

## OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

nested ended

Incrementing the value of i by 1 we get, **i = 1** and **i < 2** is true.

So, **i = 1**

# Nested loop (contd.)

OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

nested ended

Now, j is being initialized with the value 0 **again**.

So, **i = 1 & j = 0**

# Nested loop (contd.)

## OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

nested ended

1 0

Now, i and j are being printed

So, **i = 1 & j = 0**

# Nested loop (contd.)

OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.         System.out.println("nested ended");
8.     }
9. }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

nested ended

1 0

We've reached the end of the *inner loop* we have to move up again and increment j then check condition.

So, **i = 1 & j = 0**

# Nested loop (contd.)

## OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

nested ended

1 0

Incrementing the value of j by 1 we get, **j = 1** and **j < 3** is true.

So, **i = 1 & j = 1**

# Nested loop (contd.)

OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

nested ended

1 0

1 1

Now, i and j are being printed

So, **i = 1 & j = 1**

# Nested loop (contd.)

OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.         System.out.println("nested ended");
8.     }
9. }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

nested ended

1 0

1 1

We've reached the end of the *inner loop* we have to move up again and increment j then check condition.

So, **i = 1 & j = 1**

# Nested loop (contd.)

OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

nested ended

1 0

1 1

Incrementing the value of j by 1 we get, **j = 2** and **j < 3** is true.

So, **i = 1 & j = 2**

# Nested loop (contd.)

OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

nested ended

1 0

1 1

1 2

Now, i and j are being printed

So, **i = 1 & j = 2**

# Nested loop (contd.)

OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.         System.out.println("nested ended");
8.     }
9. }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

nested ended

1 0

1 1

1 2

We've reached the end of the *inner loop* we have to move up again and increment j then check condition.

So, **i = 1 & j = 2**

# Nested loop (contd.)

OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

nested ended

1 0

1 1

1 2

Incrementing the value of j by 1 we get, **j = 3** and **j < 3** is **false**.

Now, the nested loop will end and **j will be destroyed**.

So, **i = 1 & j = 3**

# Nested loop (contd.)

OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

nested ended

1 0

1 1

1 2

nested ended

Now, the print statement will be executed.

# Nested loop (contd.)

OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

nested ended

1 0

1 1

1 2

nested ended

We've reached the end of the *outer loop* we have to move up again and increment *i* then check condition.  
So, **i = 1**

# Nested loop (contd.)

## OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

nested ended

1 0

1 1

1 2

nested ended

Incrementing the value of i by 1 we get, **i = 2** and **i < 2** is **false**.

So, **i=2**

# Nested loop (contd.)

## OUTPUT

```
1. public class NestedLoop1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 2 ; i++ ) {
4.             for ( int j = 0 ; j < 3 ; j++ ) {
5.                 System.out.println(i+" "+j);
6.             }
7.             System.out.println("nested ended");
8.         }
9.     }
10. }
```

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> run NestedLoop1

0 0

0 1

0 2

nested ended

1 0

1 1

1 2

nested ended

Now we exit the main method and the program ends.

# Nested loop (contd.)

- We can use this nested loop concept with **print()** and **println()** to print various patterns as output.

```
1. public class NestedLoopPattern1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 3 ; i++ ) {
4.             for ( int j = 0 ; j < 5 ; j++ ) {
5.                 System.out.print("* ");
6.             }
7.             System.out.println();
8.         }
9.     }
10. }
```

OUTPUT

```
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> run NestedLoopPattern1

* * * * *
* * * * *
* * * * *
```

# Nested loop (contd.)

- We can use this nested loop concept with **print()** and **println()** to print various patterns as output.

```
1. public class NestedLoopPattern2{
2.     public static void main(String[] args) {
3.         for ( int i = 1 ; i <= 3 ; i++ ) {
4.             for ( int j = 1 ; j <= 3 ; j++ ) {
5.                 System.out.print(j);
6.             }
7.             System.out.println();
8.         }
9.     }
10. }
```

## OUTPUT

```
Welcome to DrJava. Working directory is /home/Java110
> run NestedLoopPattern2

1 2 3
1 2 3
1 2 3
```

# Nested loop (contd.)

- We can use this nested loop concept with **print()** and **println()** to print various patterns as output. Also, we can utilize **conditional statements** as well.

```
1. public class NestedLoopPattern3{
2.     public static void main(String[] args) {
3.         for ( int i = 1 ; i <= 3 ; i++ ) {
4.             for ( int j = 1 ; j <= 3 ; j++ ) {
5.                 if ( i <= j ){
6.                     System.out.print(j);
7.                 }
8.             }
9.             System.out.println();
10.        }
11.    }
12. }
```

## OUTPUT

```
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> run NestedLoopPattern3
123
23
3
```

**THANK YOU!**